

Cooling Tower Optimisation

Task 6 (Group Report)



**Pierre BEDRUNE, Martin HARRAMBAT, Clémence-Philomène HINOT,
John HOARAU**

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Abstract

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Acronyms

BFGS Broyden-Fletcher-Goldfarb-Shanno

PSO Particle Swarm Optimisation

SA Simulated Annealing

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1 Introduction

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2 Cooling Tower

Objective: Minimising the surface of the cooling surface

Problem information:

- Design variables:
 - fixed volume
 - r_0 to r_m
- Constraint:
 - hyperboloid
- 8 simulations:
 - 1 imposed
 - x simulations with radii fixed and heights which vary
 - x simulations with height fixed and radii which vary
 - x simulations both heights and radii vary
- run 3 algorithms on each simulation: PSO, SA, and BFGS

Other information:

- hyperbolic regression with 3 points: $r_0, r_{m/2}, r_m$
- new constraint: $r_m i$ need to be greater than $r_{m/2}$ and to be lower than r_m
- usually $r_0 < r_m$
- r_0 and r_m fixed arbitrary

- calculate the volume at each step of each algorithm
- to ensure the respect of the fixed volume we will use a Penalty function
- continuous problem

3 Algorithms

3.1 Particle Swarm Optimisation method

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3.2 Simulated Annealing method

3.2.1 Principle

Simulated Annealing is an optimisation method inspired by the annealing process in metallurgy [task3_chenonceau]. The algorithm explores the search space by accepting neighbouring solutions according to a probabilistic criterion: improvements are always accepted, while degradations are accepted with a probability that decreases over time (decreasing temperature). This acceptance mechanism helps escape local minima at the beginning, then allows the algorithm to converge towards a global minimum [task3_chenonceau].

3.2.2 Acceptance Criterion (Metropolis)

The probability of accepting a worse solution is given by:

$$P(\text{acceptance}) = e^{-\frac{E_{\text{new}} - E_{\text{actual}}}{T}} \quad (1)$$

where E_{actual} is the current energy, E_{new} is the energy of the new solution, and T is the temperature parameter [task3_chenonceau].

3.2.3 Cooling Schedule

At each iteration, the temperature decreases according to:

$$T = T \times \alpha \quad (2)$$

where α is the cooling rate [task3_chenonceau].

3.2.4 Key Parameters

The SA algorithm requires the following parameters:

- T_{initial} : Initial temperature (controls exploration intensity)
- α (**cooling_rate**): Cooling speed (typically 0.95–0.999)
- T_{min} : Stopping temperature

3.2.5 Multi-Objective Strategy (Weight Scalarisation)

SA is inherently a mono-objective algorithm. To solve bi-objective problems, a scalarisation approach transforms the problem into a single objective function using a weighted sum:

$$E(x) = w_1 \cdot f_1(x) + w_2 \cdot f_2(x) + \sum_i \lambda_i \cdot \max(0, g_i(x)) \quad (3)$$

where $w_1 + w_2 = 1$ and λ_i are penalty coefficients for constraint violations. To approximate a Pareto front, multiple independent runs are performed with weights w_1 uniformly distributed between 0.005 and 0.995 [task3_chenonceau].

3.3 Broyden-Fletcher-Goldfarb-Shanno method

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4 Results

4.1 Particle Swarm Optimisation results

TO DO

4.2 Simulated Annealing results

TO DO

4.3 Broyden-Fletcher-Goldfarb-Shanno results

TO DO

4.4 Comparison between the three methods

TO DO

4.5 Comparison of the methods performances

TO DO

Explanations: As with Tasks 3 and 4, one aim of this Task is to compare the performance of the numerical techniques that you have employed by, for example, counting the number of function evaluations that are involved, recording the elapsed time, etc.

5 Discussion

TO DO → Results' discussion + Reflection on the work done

6 Conclusion

TO DO

7 Appendices

Appendix A Code Listings

Appendix A.A PSO implementation

TO DO

Appendix A.B SA implementation

TO DO

Appendix A.C BFGS implementation

TO DO

Appendix B Detailed Results Data

Appendix B.A PSO results

TO DO

Appendix B.B SA results

TO DO

Appendix B.C BFGS results

TO DO